Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *NLLB* | ***Context:*** *Amhara Region Agricultural and Microfinance Translation Cohort*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark evaluates text outputs only [Q5] using sentence-level chrF++ and spBLEU [Q455, Q645, Q648] plus 5-point XSTS [Q654]. The deployer confirmed text-only prose output is acceptable for this NLLB deployment [elicitation A4], and this matches the benchmark's output form. However, no document-level coherence, terminology-consistency, or structural-preservation metrics are reported, all evaluation is sentence-level on general-domain content [Q5, Q336], and aggregate sentence-level chrF++ improvements may not indicate fitness for binding administrative documents requiring consistent rendering across clauses. National Ethiopia adult literacy is ~51.8% [WEB-5] with rural Amhara likely lower, meaning translated documents are often relayed orally through cooperative leaders — a use case that further increases the importance of consistent terminology, which sentence-level scoring does not assess. OF was rated LOWER priority by elicitation; given form alignment at the modality level, a moderately high score is warranted, with the residual concern being absence of document-level scoring.

ID	Evidence
Q5	“we evaluated the performance of over 40,000 different translation directions using a human-translated benchmark, Flores-200, and combined human evaluation with a novel toxicity benchmark covering all languages in Flores-200 to assess translation safety.” (p.1)
Q336	“We look for best performance on the Flores-200 development set, and report detokenized BLEU on Flores-200 devtest.” (p.41)
Q455	“We use the chrF++ metric to compare the model performance (see Section 7.1).” (p.53)
Q645	“Goyal et al. (2022) propose spBLEU, a BLEU metric based on a standardized SentencePiece model (SPM) covering 101 languages, released with Flores101. In this work, we provide SPM-200 along with Flores-200 to enable measurement of spBLEU.” (p.73)
Q648	“In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate.” (p.73)
Q654	“XSTS rates each source sentence and its machine translation on a five-point scale, where 1 is the lowest score and 5 is the highest score.” (p.74)
WEB-5	Ethiopia adult literacy ~51.8% (UNESCO via World Bank, 2017) — establishes oral-relay relevance for output use (https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=ET)

Strengths

- Text-only prose output [Q5] aligns directly with the deployer's confirmed acceptable format [elicitation A4].
- chrF++ as primary metric [Q648] with documented +0.5 detectability threshold [Q624, Q631] provides interpretable signal.
- Use of standardized SPM-200 for spBLEU [Q645] supports cross-system comparability.

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Q5	“we evaluated the performance of over 40,000 different translation directions using a human-translated benchmark, Flores-200, and combined human evaluation with a novel toxicity benchmark covering all languages in Flores-200 to assess translation safety.” (p.1)
Q624	“We investigate the relationship between chrF++ score and human evaluation score to understand what quantity of automatic metric improvement in a model ablation overall, we find that chrF++ improvements of +0.5 chrF++ usually correlate to statistically significant human evaluation improvements, with a score of +1 chrF++ almost always being detectable by human evaluators.” (p.70)

Q631	“In conclusion, we believe that based on our human evaluation studies of model ablations, that an improvement of +0.5 chrF++ is often detectable by human evaluators.” (p.71)
Q645	“Goyal et al. (2022) propose spBLEU, a BLEU metric based on a standardized SentencePiece model (SPM) covering 101 languages, released with Flores101. In this work, we provide SPM-200 along with Flores-200 to enable measurement of spBLEU.” (p.73)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text prose) matches the deployer's text-only acceptance [elicitation A4]. Both benchmark and deployment operate on Amharic text in Ethiopic script [Q79, Q382].

OF-2: Text-to-speech is not in scope for the benchmark or this deployment, but oral relay through cooperative leaders is a documented end-use pattern [regional context]; the benchmark does not evaluate any output that supports oral-relay quality (e.g., terminology consistency across read-aloud clauses).

OF-3: National adult literacy ~51.8% [WEB-5], with rural Amhara expected lower — translated prose will frequently be read aloud by cooperative leaders to lower-literacy borrowers, which depends on consistent administrative vocabulary not assessed by sentence-level chrF++ [Q455, Q648]. This is a partial form-mismatch insofar as the benchmark's evaluation unit (sentence) does not reflect the deployment's accessibility-mediated use unit (document read aloud).

OF-4: Form mismatch at the modality level is minimal; the principal residual gap is the absence of document-level/terminology-consistency scoring [Q5, Q336], which constrains how reliably aggregate scores predict fitness for binding documents.

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Q5	“we evaluated the performance of over 40,000 different translation directions using a human-translated benchmark, Flores-200, and combined human evaluation with a novel toxicity benchmark covering all languages in Flores-200 to assess translation safety.” (p.1)
Q79	“We represent each language with a BCP 47 tag sequence using a three-letter ISO 639-3 code as the base subtag, which we complement with ISO 15924 script subtags, as we collected resources for several languages in more than one script.” (p.17)
Q336	“We look for best performance on the Flores-200 development set, and report detokenized BLEU on Flores-200 devtest.” (p.41)
Q382	“We then trained an encoder for Amharic and Tigrinya only, paired with English as in all our experiments, and a specific 8k SentencePiece vocabulary to better support the Ge'ez script.” (p.45)
Q455	“We use the chrF++ metric to compare the model performance (see Section 7.1).” (p.53)
Q648	“In this work, we primarily evaluate using chrF++ using the settings from sacrebleu. However, when comparing to other published works, we utilize BLEU and spBLEU where appropriate.” (p.73)
WEB-5	Ethiopia adult literacy ~51.8% (UNESCO via World Bank, 2017) — establishes oral-relay relevance for output use (https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=ET)

Information Gaps

- Amhara-specific rural literacy rate not publicly disaggregated [registry NOT FOUND].

Remediation

Gap 1: All evaluation is sentence-level [Q5, Q336]; no document-level coherence or terminology-consistency scoring, which matters for oral-relay use to lower-literacy borrowers [WEB-5].

Recommendation: Add document-level evaluation: score multi-clause document translations on (a) cross-clause terminology consistency for defined terms, (b) preservation of numbered-clause structure in prose form, and (c) intelligibility when read aloud (oral-relay test with cooperative extension worker raters).

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